

AHAAN SONI

Backend Engineer | System Design | Distributed Systems

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SUMMARY

Results-driven Backend Engineer with 5+ years of experience architecting scalable, high-throughput systems across Banking, IoT, Video Telematics, InsurTech, and EdTech. Specializes in distributed systems design, event-driven architectures (Kafka, RabbitMQ), microservices, and configuration-driven platforms. Proven track record of owning end-to-end delivery from system design through production rollout and UAT in mission-critical environments serving millions of customers.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python

Backend: Node.js, NestJS, Express.js, routing-controllers, TypeDI, REST APIs, WebSockets, Serverless Framework

Databases: PostgreSQL, MongoDB, DynamoDB, Redis, Pinecone, TypeORM, Sequelize

Messaging & Streaming: Apache Kafka, RabbitMQ, MQTT, Protobuf

Architecture & Patterns: Microservices, SAGA, Circuit Breaker, Event-Driven Design, STP / Config-Driven Flows, OOP

Cloud & DevOps: AWS (EC2, Lambda, API Gateway, S3, CloudWatch, IAM, Elastic Beanstalk), Docker, Jenkins, CI/CD

Observability: Kibana, Grafana, Prometheus, AppDynamics

Testing & Tooling: Jest, ts-jest,nock, mock-req-res, Swagger/OpenAPI, Postman, ESLint, Prettier, Husky, Git, JIRA

AI-Assisted Dev: Claude Code, ENBD Leap (@enbdleap/ai-llm)

PROFESSIONAL EXPERIENCE

Nagarro

Associate Staff Engineer

Jul 2025 – Present

Bengaluru, India

Client: Emirates NBD (ENBD) CRMX-FCR | Backend platform powering IVR & customer-service agent workflows, running configuration-driven Straight-Through-Processing (STP) flows across multiple core banking systems.

- Owned end-to-end delivery of a new STP for Skywards miles cancellation from config schema design and automation to data-load testing and JIRA closure integrating with ENBD's loyalty backend.
- Re-engineered the Contact Details Update (Paperless Journey) submission pipeline to support multiple addresses and contact fields with varying schemas entirely config-driven, eliminating code changes for future field additions; authored 1900+ lines of DB configuration and shipped in ~2 weeks.
- Unblocked failing deferment cases in production via targeted config-driven fixes (zero code change) and extended the API response contract to expose the deferment field for the frontend.
- Built an automated test suite for Customer Service flows using Claude Code and AI-assisted tooling; attached coverage reports to JIRA tickets for traceability.
- Provided UAT and war-room support for the Early Settlement STP rollout diagnosing live issues across orchestration, validation, and core banking integration layers.
- Authored CLAUDE.md repo-context documentation describing architecture, bootstrap loaders, and service patterns to onboard team members and improve AI-assisted development across the codebase.
- Tech: Node.js, TypeScript, Express + routing-controllers, TypeDI, microframework-w3tec, MongoDB, PostgreSQL, Kafka (SASL), Redis, SAGA, Circuit Breaker, ENBD Leap framework, Jest, Swagger, Jenkins, Kibana, Grafana, AppDynamics.

Intangles

Backend Developer

Nov 2024 – June 2025

Pune, India

Camera-Based Driver Intelligence System

- Built a real-time driver monitoring pipeline processing data from 15,000+ cameras every 2 minutes, using Kafka for stream ingestion and scalable batch processing.
- Engineered dynamic driver intelligence records by detecting and validating driver identity per vehicle run span (daily 12 AM–12 AM), ensuring accurate driving-hour tracking for payroll and compliance.
- Implemented validation logic to exclude ambiguous entries, significantly improving the trustworthiness of driver-hour records used across logistics operations.

Configuration Over-the-Air (COTA) System

- Designed and implemented a scalable COTA backend to manage 250+ camera configuration settings across 10,000+ edge devices using MQTT and RabbitMQ, enabling real-time remote updates without manual intervention.
- Solved critical sync issues between camera state and DB configurations, ensuring zero-lag propagation of updated settings to all devices.
- Replaced heavy JSON-based config comparisons with compact Protobuf serialization reducing payload size by ~70% and significantly improving comparison speed.

ADAS / DMS Alert Management System

- Developed a scalable Python-based consumer processing 80+ camera messages per second for ADAS and Driver Monitoring Systems, generating 12 distinct real-time safety alerts.
- Integrated AI-driven detection algorithms and built a parallel alert-handling pipeline using RabbitMQ to fetch 10-second incident video clips from Amazon S3, reducing processing latency by 40%.
- Refactored legacy components to enable asynchronous, multi-threaded video retrieval and alert publishing, improving throughput and reliability under high load.

Gemini Solutions

Senior Software Engineer 1 → Software Engineer 2 → Software Engineer 1

Mar 2022 – Nov 2024

Gurgaon, India

Emaar Group — B2B Mall Services Dashboard

- Transformed Emaar's B2B client experience by developing an interactive dashboard for managing services, requests, and 1,000+ shops across their global mall portfolio.

TATA Banca — Pan-India Insurance Platform

- Led development of a Pan-India insurance web app using microservices and serverless architecture; boosted user engagement by 30% and conversion by 25%, managing over 36,000 CR policies across 10+ partner banks.

Project Untold — Cinema Investment Platform

- Led development of an innovative cinema investment platform on the MERN stack owning architecture, API design, and core transaction flows.

Athletic Estate — Merchandise Platform

- Built a merchandise sales platform using React, Node.js, PostgreSQL, and Tailwind, increasing sales by 25%.

Schoollog

Backend Developer

Oct 2021 – Feb 2022

Jaipur, India

- Delivered ERP features end-to-end for school management under senior guidance.
- Bridged client engagement with technical implementation for the COCO project.

EDUCATION

Bachelor of Computer Application

Lal Bahadur Shastri P.G. College

Jul 2014 – Aug 2017

Jaipur, India